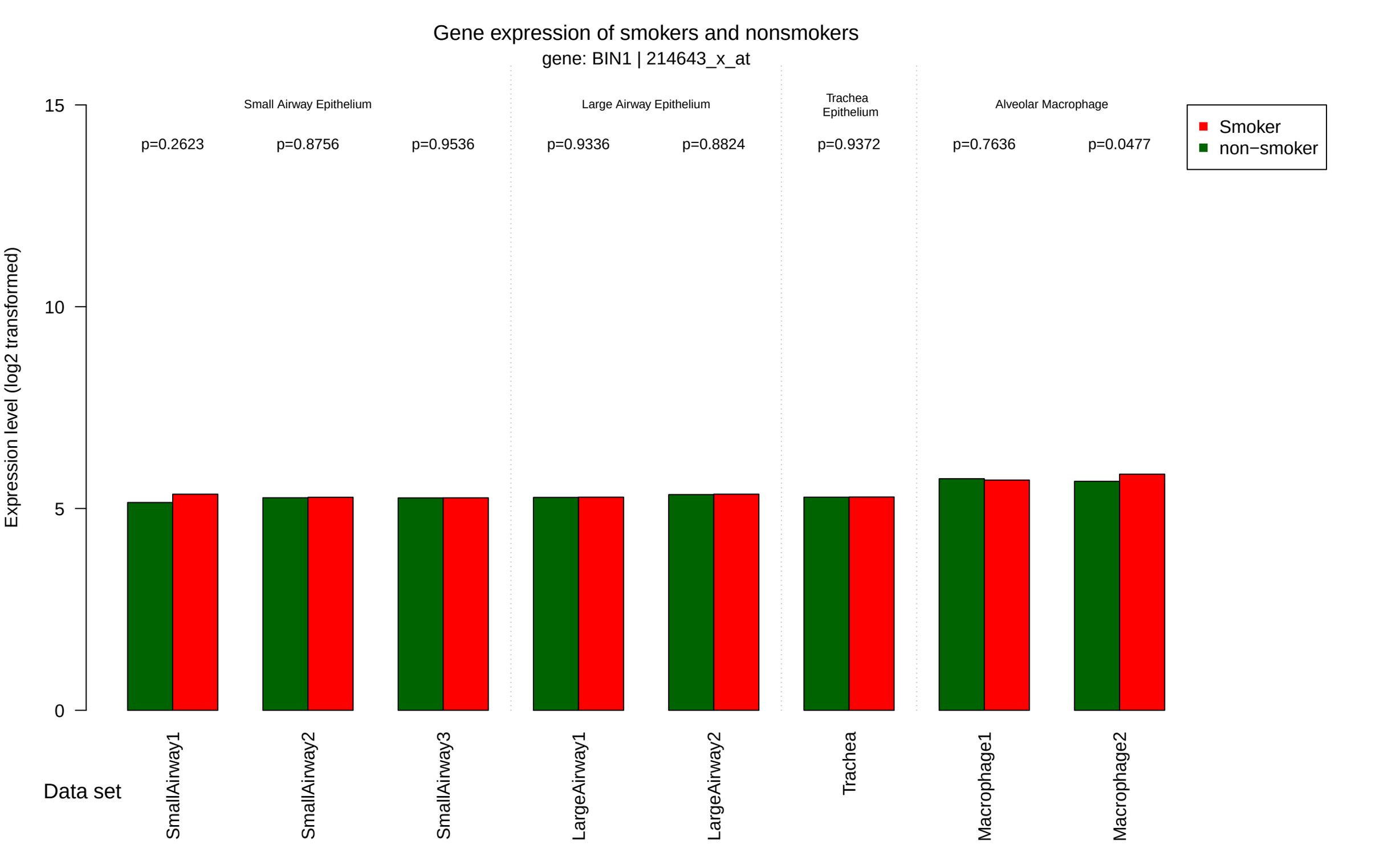
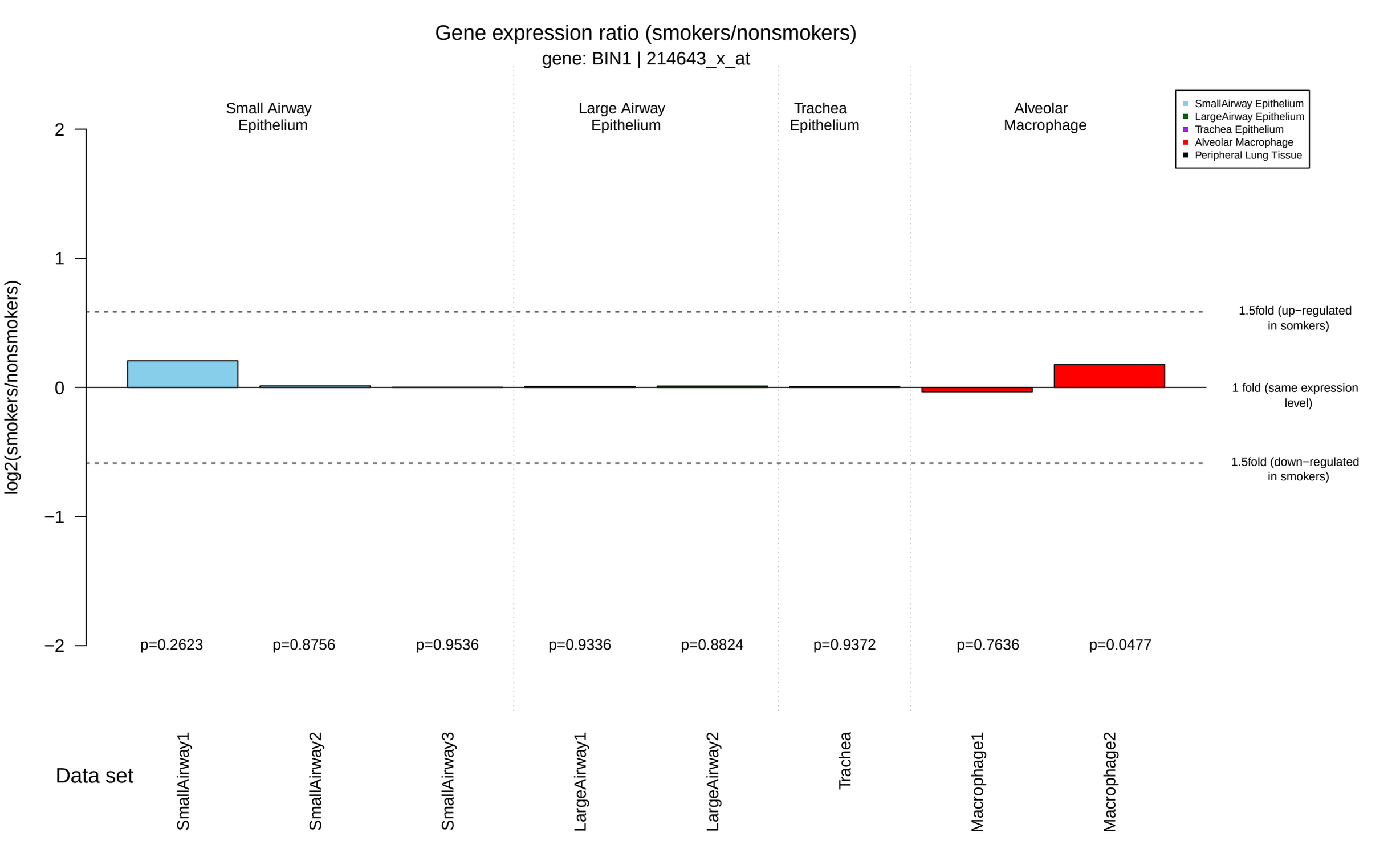
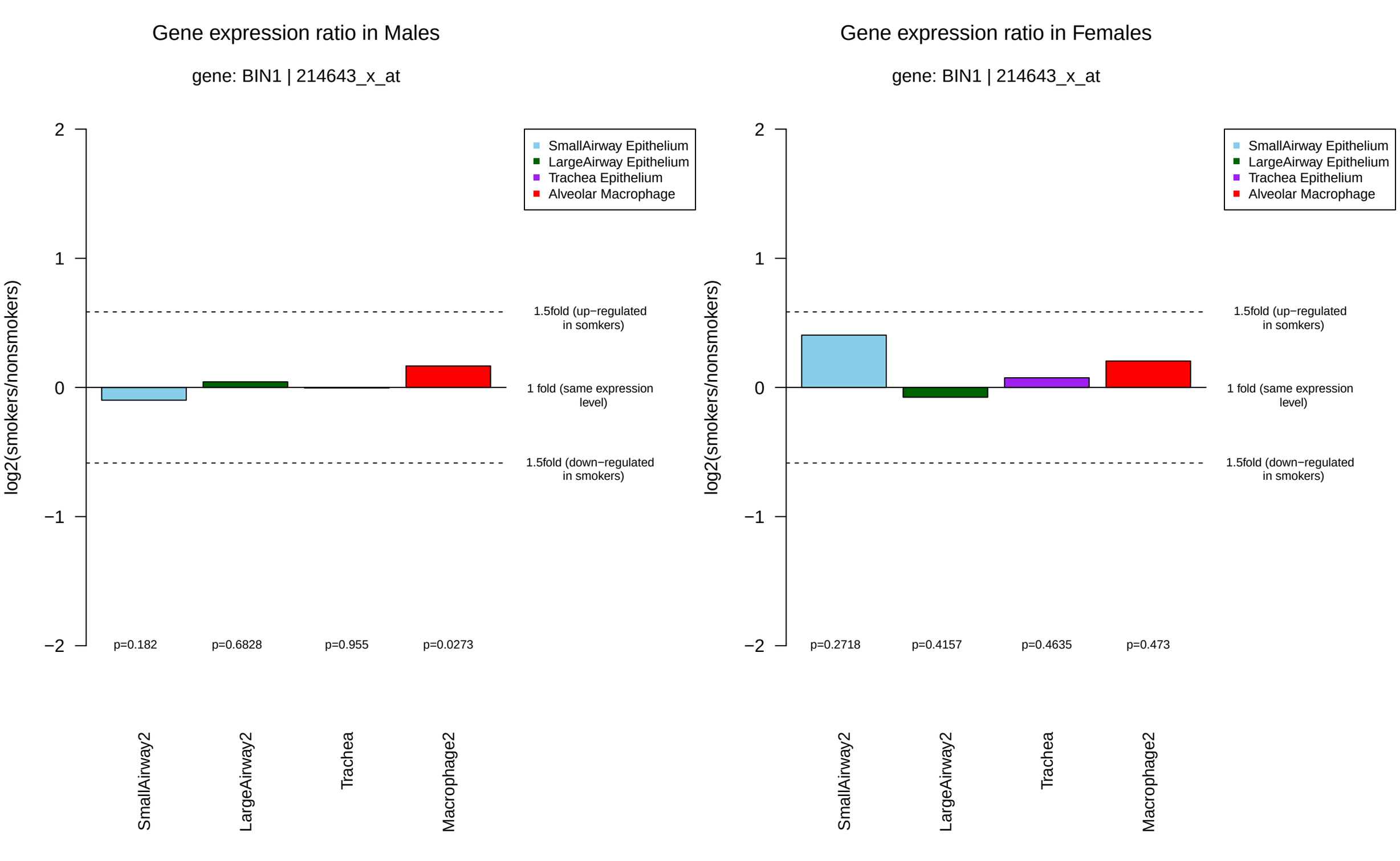
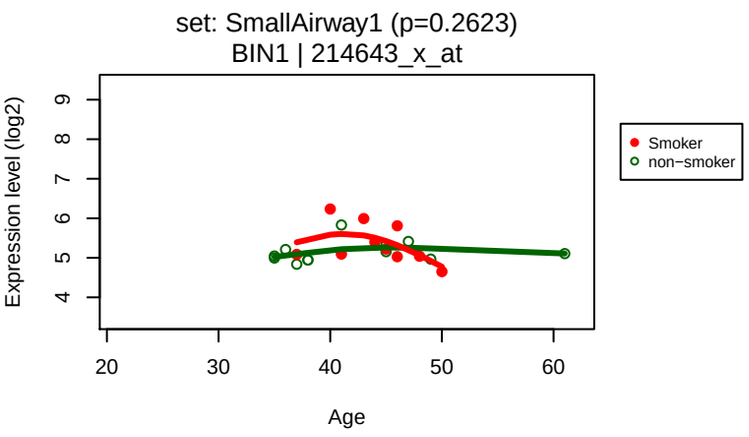
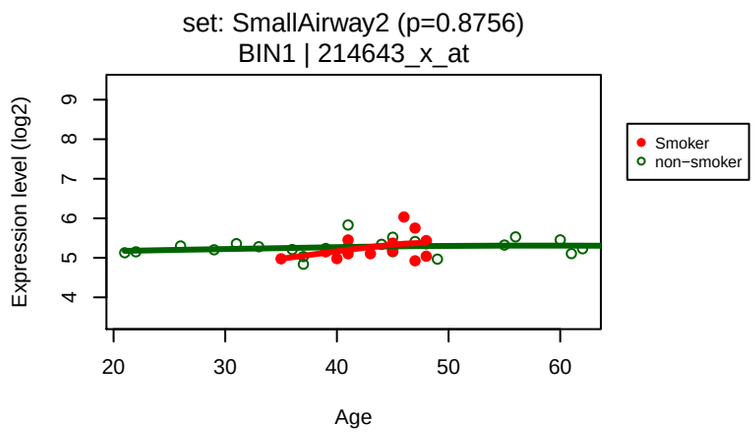
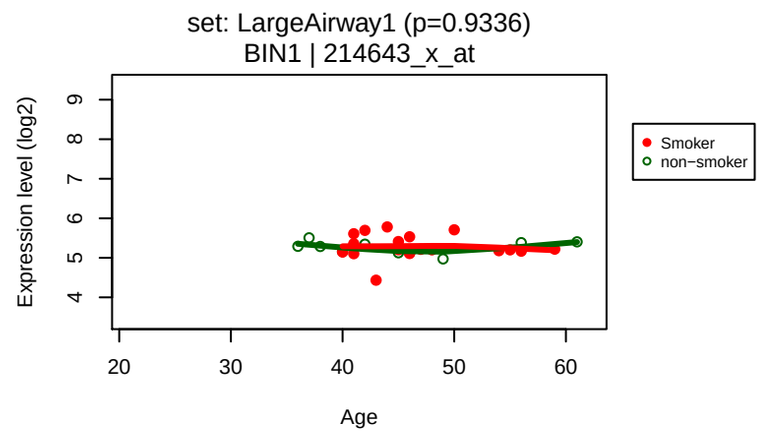

Gene expression of smokers and nonsmokers
gene: BIN1 | 214643_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2623
p=0.8756
p=0.9536
p=0.9336
p=0.8824
p=0.9372
p=0.7636
p=0.0477
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

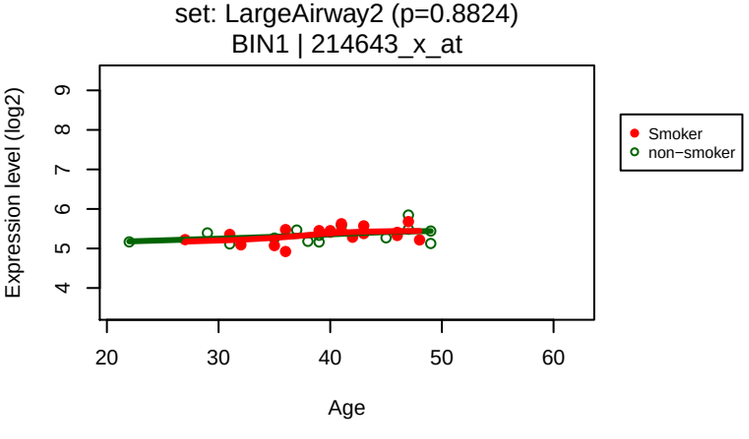

Gene expression ratio (smokers/nonsmokers)
gene: BIN1 | 214643_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2623
p=0.8756
p=0.9536
p=0.9336
p=0.8824
p=0.9372
p=0.7636
p=0.0477
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

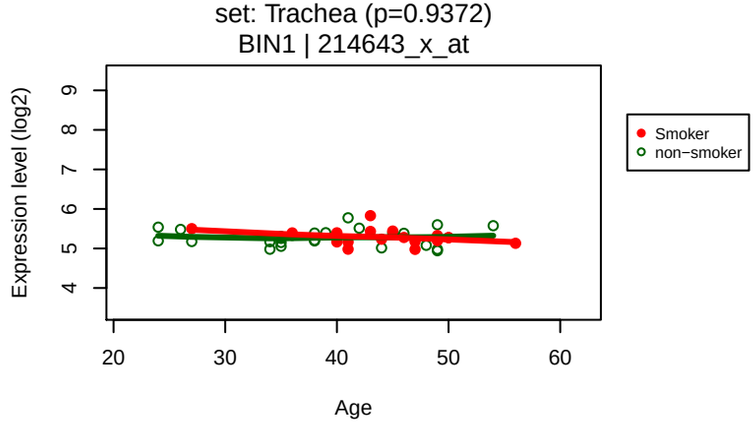

Gene expression ratio in Males
gene: BIN1 | 214643_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.182
p=0.6828
p=0.955
p=0.0273
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BIN1 | 214643_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2718
p=0.4157
p=0.4635
p=0.473
SmallAirway2
LargeAirway2
Trachea
Macrophage2

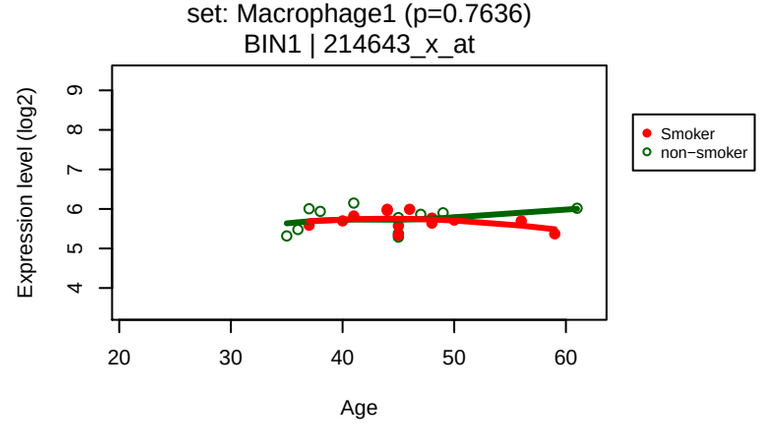

set: SmallAirway1 (p=0.2623)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker

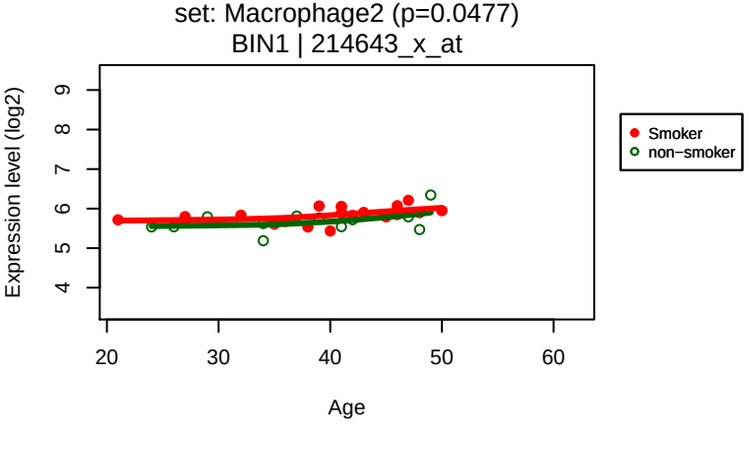

set: SmallAirway2 (p=0.8756)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker

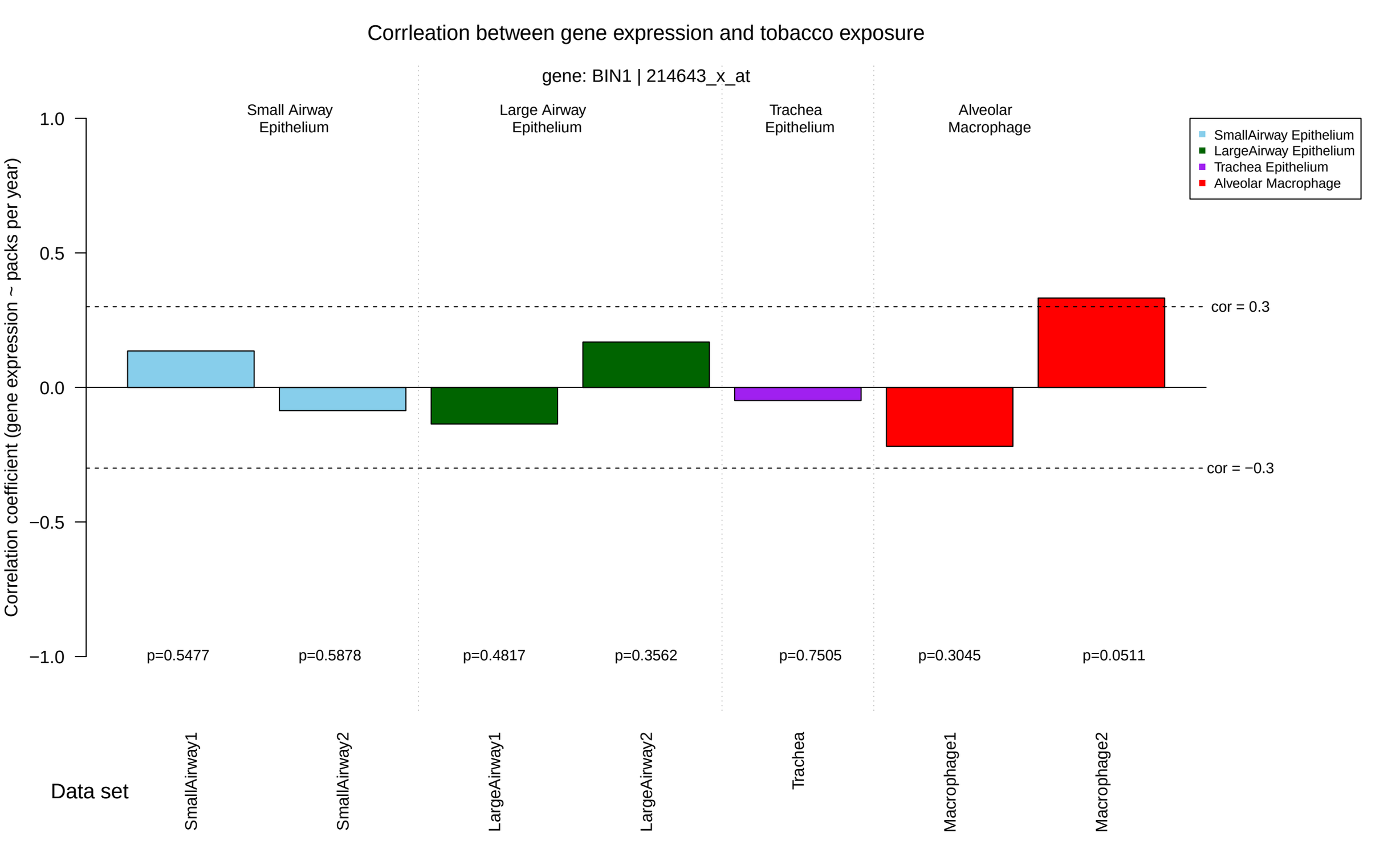

set: LargeAirway1 (p=0.9336)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8824)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9372)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7636)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0477)
BIN1 | 214643_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BIN1 | 214643_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5477
p=0.5878
p=0.4817
p=0.3562
p=0.7505
p=0.3045
p=0.0511
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2